

T0-Theory: Cosmic Relations

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Abstract

The T0-theory examines how far cosmic phenomena can be traced back to a single universal constant $\xi = \frac{4}{3} \times 10^{-4}$. This document presents the fundamental relationships between the gravitational constant, cosmic microwave background radiation (CMB), Casimir effect and cosmic structures within the framework of a static, eternally existing universe. All derivations are performed in natural units ($\hbar = c = k_B = 1$) and respect the time-energy duality as a fundamental principle of quantum mechanics.

Note on versions. The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus. This document deals almost exclusively with the cosmic sector (static universe, CMB, redshift, dark energy). The current corpus

deliberately leaves this sector aside, because the Standard Model does not derive it either (P39); the cosmological conclusions of this document are therefore to be read as **[S]**.

.1 Introduction: The Universal ξ -Constant

Foundations of T0 Theory

Important

T0 theory is based on the universal dimensionless constant $\xi = \frac{4}{3} \times 10^{-4}$, to which it traces phenomena from the subatomic to the cosmic scale.

T0 theory is built on a single fundamental constant. This constant forms the basis for all physical calculations and predictions of the theory:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333333... \times 10^{-4} \quad (1)$$

This dimensionless constant connects quantum and gravitational phenomena and is intended as the basis of a unified description of the fundamental interactions.

Note on Derivation

For the detailed derivation and physical justification of this fundamental constant, see the document "Parameter Derivation" (available at:).

Time-Energy Duality as Foundation

Thesis

[S] From Heisenberg's uncertainty relation $\Delta E \times \Delta t \geq \hbar/2 = 1/2$ (natural units) this document concludes that a classical Big Bang singularity with infinite density is not physically realized and is replaced by a tiny but finite core with minimal length scale L_0 (from ξ).

Heisenberg's uncertainty relation between energy and time represents the fundamental principle of T0-theory:

$$\Delta E \times \Delta t \geq \frac{1}{2} \quad (\text{natural units}) \quad (2)$$

In the reading of this document, this relation has far-reaching cosmological consequences **[S]**:

- A temporal beginning (Big Bang) would mean $\Delta t = \text{finite}$
- This leads to $\Delta E \rightarrow \infty$ - physically inconsistent
- Therefore the universe must have existed eternally: $\Delta t = \infty$
- The universe is static, without expanding space

.2 Cosmic Microwave Background (CMB)

CMB without Big Bang: ξ -Field Mechanisms

Thesis

[S] If time-energy duality excludes a Big Bang, as concluded above, the CMB must have a different origin than the $z=1100$ decoupling of standard cosmology.

T0-theory interprets the CMB **[S]** as radiation from ξ -field quantum fluctuations:

$$\frac{T_{\text{CMB}}}{E_{\xi}} = \frac{16}{9} \xi^2 \quad (3)$$

With $E_{\xi} = \frac{1}{\xi} = \frac{3}{4} \times 10^4$ (natural units) and $\xi = \frac{4}{3} \times 10^{-4}$ this yields:

$$T_{\text{CMB}} = \frac{16}{9} \xi^2 \times E_{\xi} = \frac{16}{9} \times 1.78 \times 10^{-8} \times 7500 = 2.35 \times 10^{-4} \quad (4)$$

Conversion to SI units:

$$T_{\text{CMB}} = 2.725 \text{ K} \quad (5)$$

The expression thus lies about 1 % from the observed value (cf. the table of dimensionless ratios: 3.13 versus 3.16×10^{-8}).

CMB Energy Density and ξ -Length Scale

The CMB energy density in natural units is:

$$\rho_{\text{CMB}} = 4.87 \times 10^{41} \quad (\text{natural units, dimension } [E^4]) \quad (6)$$

This energy density defines a characteristic ξ -length scale:

$$L_{\xi} = \left(\frac{\xi}{\rho_{\text{CMB}}} \right)^{1/4} \quad (7)$$

Fundamental relation of CMB energy density:

$$\rho_{\text{CMB}} = \frac{\xi}{L_{\xi}^4} = \frac{\frac{4}{3} \times 10^{-4}}{(L_{\xi})^4} \quad (8)$$

.3 Casimir Effect and ξ -Field Connection

Casimir-CMB Ratio as Consistency Check

Experiment

The ratio between Casimir energy density and CMB energy density is consistent with the characteristic ξ -length scale of $L_{\xi} = 10^{-4} \text{ m}$.

The Casimir energy density at plate separation $d = L_\xi$ is:

$$|\rho_{\text{Casimir}}| = \frac{\pi^2}{240 \times L_\xi^4} \quad (\text{natural units}) \quad (9)$$

The experimental ratio yields:

$$\frac{|\rho_{\text{Casimir}}|}{\rho_{\text{CMB}}} = \frac{\pi^2}{240\xi} = \frac{\pi^2 \times 10^4}{320} \approx 308 \quad (10)$$

Direct calculation with SI values: With $L_\xi = 10^{-4}$ m, direct calculation gives:

$$|\rho_{\text{Casimir}}| = \frac{\hbar c \pi^2}{240 \times (10^{-4})^4} = 1.3 \times 10^{-11} \text{ J/m}^3 \quad (11)$$

$$\rho_{\text{CMB}} = 4.17 \times 10^{-14} \text{ J/m}^3 \quad (12)$$

$$\text{Ratio} = \frac{1.3 \times 10^{-11}}{4.17 \times 10^{-14}} = 312 \quad (13)$$

The theoretical expression (308) and the direct calculation with the measured ρ_{CMB} (312) differ by 1.3%.

ξ -Field as Universal Vacuum

Important

The ξ -field manifests both in free CMB radiation and in geometrically constrained Casimir vacuum. In the interpretation of this document this is a hint at the physical reality of the ξ -field **[S]**; it is not a proof.

The characteristic ξ -length scale L_ξ is the point where CMB vacuum energy density and Casimir energy density reach comparable magnitudes:

$$\text{Free vacuum: } \rho_{\text{CMB}} = +4.87 \times 10^{41} \quad (14)$$

$$\text{Constrained vacuum: } |\rho_{\text{Casimir}}| = \frac{\pi^2}{240d^4} \quad (15)$$

.4 Cosmic Redshift without Expansion

ξ -Field Energy Loss Mechanism

Thesis

[S] According to this thesis, the observed cosmic redshift arises not from spatial expansion but from energy loss of photons in the omnipresent ξ -field. The wavelength dependence assumed here has been replaced in the current corpus by an achromatic redshift (correction K6 in Doc. 190).

Photons lose energy through interaction with the ξ -field:

$$\frac{dE}{dx} = -\xi \cdot f\left(\frac{E}{E_\xi}\right) \cdot E \quad (16)$$

For the linear case $f\left(\frac{E}{E_\xi}\right) = \frac{E}{E_\xi}$ this yields:

$$\frac{dE}{dx} = -\frac{\xi E^2}{E_\xi} \quad (17)$$

Wavelength-Dependent Redshift

Integration of the energy loss equation leads to wavelength-dependent redshift:

Wavelength-dependent redshift:

$$z(\lambda_0) = \frac{\xi x}{E_\xi} \cdot \lambda_0 \quad (18)$$

where λ_0 is the emitted wavelength and x is the distance traveled.

This formula predicts:

- Shorter wavelength light (UV) shows greater redshift
- Longer wavelength light (radio) shows smaller redshift
- The ratio is $z_1/z_2 = \lambda_1/\lambda_2$

Experiment

Experimental test: Comparison of radio and optical redshifts

- 21cm hydrogen line: $\nu = 1420$ MHz
- Optical H α line: $\nu = 457$ THz
- Predicted ratio: $z_{21\text{cm}}/z_{\text{H}\alpha} = 3.1 \times 10^{-6}$

.5 Structure Formation in the Static ξ -Universe

Continuous Structure Development

In the static T0 universe **[S]**, structure formation occurs continuously without Big Bang constraints:

$$\frac{d\rho}{dt} = -\nabla \cdot (\rho \mathbf{v}) + S_\xi(\rho, T, \xi) \quad (19)$$

where S_ξ is the ξ -field source term for continuous matter/energy transformation.

ξ -Supported Continuous Creation

The ξ -field enables continuous matter/energy transformation:

$$\text{Quantum vacuum} \xrightarrow{\xi} \text{Virtual particles} \quad (20)$$

$$\text{Virtual particles} \xrightarrow{\xi^2} \text{Real particles} \quad (21)$$

$$\text{Real particles} \xrightarrow{\xi^3} \text{Atomic nuclei} \quad (22)$$

$$\text{Atomic nuclei} \xrightarrow{\text{Time}} \text{Stars, galaxies} \quad (23)$$

Energy balance is maintained by:

$$\rho_{\text{total}} = \rho_{\text{matter}} + \rho_{\xi\text{-field}} = \text{constant} \quad (24)$$

.6 Dimensionless ξ -Hierarchy

Energy Scale Ratios

The ξ -relations used here reduce to exact mathematical ratios:

Table 1: Dimensionless ξ -ratios

Ratio	Expression	Value
Temperature	$\frac{T_{\text{CMB}}}{E_{\xi}}$	3.13×10^{-8}
Theory	$\frac{16}{9} \xi^2$	3.16×10^{-8}
Length	$\frac{\ell_{\xi}}{L_{\xi}}$	$\xi^{-1/4}$
Casimir-CMB	$\frac{ \rho_{\text{Casimir}} }{\rho_{\text{CMB}}}$	$\frac{\pi^2 \times 10^4}{320}$

Important

The ξ -relations used here consist of exact mathematical ratios:

- Fractions: $\frac{4}{3}, \frac{3}{4}, \frac{16}{9}$
- Powers of ten: $10^{-4}, 10^3, 10^4$
- Mathematical constants: π^2

No arbitrary decimal numbers occur in these expressions; they are built from ξ and integer structural choices (motivated settings).

.7 Experimental Predictions and Tests

Precision Measurements of Gravitational Constant

T0-theory predicts:

$$G_{\text{T0}} = 6.67430000... \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2) \quad (25)$$

This prediction can be tested by future precision measurements.

Casimir Force Anomalies

Experiment

Prediction: Casimir force anomalies at characteristic ξ -length scale

- Standard Casimir law: $F \propto d^{-4}$
- ξ -field modifications at $d = L_{\xi} = 10^{-4} \text{ m}$
- Measurable deviations through ξ -vacuum coupling

Electromagnetic Resonance

Maximum ξ -field-photon coupling at characteristic frequency:

$$\nu_{\xi} = \frac{1}{L_{\xi}} = 10^4 \text{ Hz} = 10 \text{ kHz} \quad (26)$$

Electromagnetic anomalies should occur at this frequency.

.8 Cosmological Consequences

Solution to Cosmological Problems

The following table shows how known problems of standard cosmology appear in the static T0 model **[S]**. The entries are interpretations in the cosmic sector, which the current corpus leaves aside (P39), not derived solutions:

Table 2: Cosmological problems: Standard vs. T0

Problem	Λ CDM	T0 Interpretation
Horizon problem	Inflation required	Infinite causal connectivity
Flatness problem	Fine-tuning	Geometry stabilizes over infinite time
Monopole problem	Topological defects	Defects dissipate over infinite time
Lithium problem	Nucleosynthesis discrepancy	Nucleosynthesis over unlimited time
Age problem	Objects older than universe	Objects can be arbitrarily old
H_0 tension	9% discrepancy	No H_0 in static universe
Dark energy	69% of energy density	Not required

Parameter Reduction

Parameter count

Parameter count: the Standard Model and Λ CDM together use 25+ parameters, T0-theory the single parameter ξ (plus integer structural choices as motivated settings; SI anchors serve only for conversion).

- Standard model of particle physics: 19+ parameters
- Λ CDM cosmology: 6 parameters
- T0-theory: 1 parameter (ξ)

Arithmetically this corresponds to a 96% reduction. The Λ CDM share concerns the cosmic sector, which the current corpus leaves aside (P39).

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